
Uncovering Conceptual Blindspots in Generative Image Models Using Sparse Autoencoders

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Abstract

Despite their impressive performance, generative image models trained on large-scale datasets still exhibit clear failures on seemingly simple concepts—e.g., human hands or objects appearing in groups of four—that should plausibly be represented in the training distribution. These failures have largely been treated as anecdotes, obscuring the stronger possibility that they are systematic limitations of current generative models rather than isolated quirks. We address this gap by introducing a systematic framework for identifying and characterizing "conceptual blindspots"—concepts present in the training data but absent or misrepresented in a model’s generations. Our method uses sparse autoencoders (SAEs) to extract interpretable concept embeddings and directly quantify concept prevalence gaps between real and generated images. We train an archetypal SAE (RA-SAE) on DINOv2 features with 32,000 concepts—the largest such SAE to date—and obtain the resolution needed to expose fine-grained conceptual disparities. Applied to four widely used generative models (Stable Diffusion 1.5/2.1, PixArt, and Kandinsky), the framework uncovers concrete suppressed blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated blindspots (e.g., wood background texture and palm trees). At the individual-datapoint level, the same analysis isolates memorization artifacts—instances where models reproduce highly specific visual templates seen during training. Together, these results establish a theoretically grounded and empirically actionable framework for measuring conceptual fidelity with respect to the underlying data-generating process and for systematically exposing blindspots in generative models.

1 Introduction

Generative image models trained on large-scale datasets now deliver unprecedented visual synthesis capabilities, enabling applications across vision [1–9] and far beyond it [9–15]. Yet qualitative [16–18] and quantitative studies [19–20] show that these models still fail on basic concepts, including human hands [21–24], objects appearing in groups of four [25], and negations or object relations [20, 26]. When prompted with such concepts, models often generate nearby visual structures without realizing the target concept itself—for example, producing hands with six fingers. We call these failures “conceptual blindspots”¹. They are especially important because the missing concepts are not obscure edge cases; one would expect large-scale training data to contain many accurate demonstrations of them. The central question is therefore not whether individual prompts can fail, but whether these failures reveal a systematic phenomenon: across a broad spectrum of concepts, models either under-produce or over-produce specific visual content relative to the data distribution.

¹We borrow the term “blindspots” from psychology literature [27], wherein it is used to refer to scenarios where an agent makes biased decisions despite exposure to observations that contradicts their decision’s rationale.



Figure 1: $\delta(k)$ quantifies how strongly a model over- or under-generates a concept c_k relative to its natural-data frequency. Concepts with $\delta(k) < 0.1$ are **suppressed conceptual blindspots**; concepts with $\delta(k) > 0.9$ are **exaggerated conceptual blindspots**. The images generated by four popular generative image models show both decisive blindspot cases and aligned concepts.

Existing evaluation approaches cannot directly answer this concept-level question [28, 29]. Most methods rely on coarse-grained measures of image realism, such as FID [30], and therefore miss distributional failures at the level of individual concepts. Methods such as CLIPScore, diversity metrics, and distribution-coverage statistics [31–33] provide useful but incomplete evidence: they do not resolve fine-grained features or concepts [34, 35], which is precisely the granularity required to identify conceptual blindspots. Qualitative analyses, including participant surveys [5, 17, 36–38] and open-ended exploration [39], can reveal individual failures, but they do not provide a scalable, repeatable methodology for comparing concept-level weaknesses across model families.

This work. We argue that identifying conceptual blindspots requires an automated, unsupervised methodology that pinpoints concepts whose generation odds differ between the true data-generating process and the trained model. Our contributions in this work are three-fold:

- **Formalizing Conceptual Blindspots in Generative Image Models.** We introduce a precise framework for identifying and quantifying conceptual blindspots in generative image models relative to natural images (Section 3). This formalization replaces anecdotal or human-defined evaluations with a principled basis for measuring model limitations.
- **A Scalable, Unsupervised Approach for Identifying Conceptual Blindspots using Sparse Autoencoders.** We develop a scalable methodology that uses sparse autoencoders (SAEs) to extract and compare concept distributions between natural and synthesized images (Section 4). To support this analysis, we train and open-source an archetypal SAE (RA-SAE) on DINOv2 features with 32, 000 concepts, the largest such RA-SAE to date.
- **Exploratory Tool and Analysis.** Our exploratory web tool enables distribution- and datapoint-level analysis of blindspots across models (Sections 5.1–5.4). We apply the method to Kandinsky, PixArt, and Stable Diffusion (SD) 1.5/2.1 (Section 5) and identify concrete suppressed conceptual blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated conceptual blindspots (e.g., wood background texture and palm trees), shown in Fig. 1.

The model, code, and web tool are available at a software repository.

2 Related Work

2.1 Explainability in Vision

Early work in explainable AI for computer vision centered on attributing influential input regions [40–42]. These methods were valuable but insufficient for our goal: they provided limited semantic information about learned representations and could produce incorrect explanations [43–45]. Concept-based interpretability [46] directly addresses this gap by identifying semantically meaningful directions in neural networks, revealing not only where models attend but also which concepts and structures they use [47–49].

Recent work establishes that widely used concept-based interpretability methods—ACE [44], CRAFT [48], and Sparse Autoencoders (SAEs) [50, 51]—solve closely related dictionary-learning problems under different constraints [52]. Among these approaches, SAEs provide the most scalable route to concept-based analysis. Although original SAEs have documented limitations, including overly specific features [53], compositionality challenges [54], and limited intervention effects [55],

improved variants such as archetypal SAE (RA-SAE) [56] directly strengthen the reliability of the learned concept basis.

Beyond SAEs, prompt-based probing [57], attention-map or activation visualizations [58, 59], and dataset-level statistics [32, 33] such as diversity or distribution-coverage metrics offer only partial evidence for conceptual failures. Their core limitation is granularity: they typically operate on full images or prompts rather than fine-grained features and concepts [34, 35]. They also often depend on subjective interpretation and do not cleanly separate different failure modes [60]. Consequently, existing methods and metrics do not provide the systematic concept-level diagnostic needed to expose the weaknesses of generative image models [28].

2.2 Generative Image Models

Diffusion-based methods now dominate generative vision modeling across images [4, 61–64], video [65–72], and 3D [7, 73–75]. For image generation, this trajectory begins with denoising diffusion probabilistic models (DDPMs) [76] and extends to non-Markov diffusion processes such as denoising diffusion implicit models (DDIMs) [77].

The Stable Diffusion (SD) [9] family made DDPM-based image generation broadly accessible to both research and open-source communities. Subsequent releases, including SD 2 [78], SD 3 [79], and SD XL [80], were accompanied by extensions that impose additional constraints on style [81, 82], pose [83], and identity [84, 85], as well as variants for other input modalities such as image-to-image generation. The ecosystem now includes multiple Latent Diffusion Models built on or around this paradigm—including Kandinsky [86], PixArt [87], DeepFloyd [4, 88], and FLUX [89]—making cross-model conceptual diagnostics increasingly necessary.

2.3 Datasets for Generative Image Models

The success of modern generative vision models rests on large-scale compute and internet-scale datasets [90–92]. LAION-5B [93] is especially central for open-source text-to-image models, including SD and its derivatives. Scraped from Common Crawl [94], it contains more than 5B image-caption pairs, including 2.3B in English. Other major datasets, such as COYO-700M [95] and Conceptual Captions [96], provide 700M and 3M image-caption pairs, respectively.

As LAION-5B became widely used, its biases and safety issues became equally salient [97–101]. Filtering did not fully remove harmful content [97–99], including NSFW material [100] and hundreds of CSAM instances [101], which prompted temporary removal from official channels. The dataset also contains low-quality images [102] and internet-style captions such as product descriptions, creating a measurable mismatch with how users prompt trained models [103].

2.4 Concept Discovery and Sparse Coding in Generative Image Models

Dictionary learning provides the technical foundation for representing data through sparse combinations of learned atoms [104–106]. Built on compressed sensing theory [107, 108], the field has progressed from vector quantization [109] to methods such as Non-negative Matrix Factorization [110, 111], Sparse PCA [112], and K-SVD [113]. Online methods [114], structured sparsity [115], and theoretical guarantees [116, 117] now connect this tradition directly to deep representation learning [118, 119].

Sparse coding has already proved useful for studying the emergence of high-level concepts inside diffusion models [120]. Earlier concept-grounded approaches for generative architectures relied on concept-bottleneck models requiring human intervention during training [121] or post-hoc detectors that retrofit concept supervision [122]. Both approaches remain limited because they require human-defined concepts and therefore miss broader conceptual trends that users have not explicitly specified.

Aggregate metrics such as precision, recall, density, and coverage [35, 123], along with latent density scores that estimate sample quality from the model’s latent space [124], evaluate important aspects of generative capability. They can expose distributional gaps, but they do not identify which specific concepts are under- or over-represented.

These limitations make a scalable, unsupervised framework for identifying and quantifying concept-level failure modes in generative image models necessary [125].

3 Formalizing Conceptual Blindspots in Generative Models

We formalize *conceptual blindspots* as systematic discrepancies between the conceptual content of natural images and the conceptual content of model-generated outputs. This formulation yields principled quantitative measures that determine which concepts a generative model suppresses or exaggerates relative to its data distribution. Fig. 2 summarizes the resulting process. The formalization uses standard assumptions [126–130], and our experiments show that the resulting phenomenology remains informative even when those assumptions are only approximate.

Definition 1 (Data-Generating Process). Let $\mathcal{C} \subset \mathbb{R}^K$ denote a latent space with a Boltzmann prior $p(c) = \exp(-E(c))Z^{-1}$, where $E(\cdot)$ denotes an energy function that linearly decomposes over individual latents and Z is the corresponding partition function, i.e., $E(c) = \sum_k E(c_k)$ and hence $p(c) = \prod_k p_k(c_k)$. A data-generating process (DGP) is an invertible function $G : \mathcal{C} \rightarrow \mathcal{X}$ that maps the latents $c \in \mathcal{C}$ to observations $x \in \mathcal{X}$, i.e., $x = G(c)$.

For notational simplicity, we use $p(\cdot)$ to denote both the latent density $p(c)$ and its push-forward to image space $p(x)$, where $x = G(c)$. This is justified by the invertibility of G , which induces a valid distribution over $\mathcal{X}$ via the change of variables formula. The individual dimensions of the latent space serve as the **Concepts** underlying the data distribution $P_{\mathcal{X}}$ over the image observation space $\mathcal{X}$. For example, different latents can correspond to concepts such as color, shape, size, location, and related visual attributes [131, 132]. We allow the data-generating process to associate a text description $t \in \mathcal{T}$ with each sampled image, although we do not explicitly model this text-generation mechanism. These descriptions train a text-conditioned **Generative image model** g_{θ} , with parameters θ , to map a noise signal $\eta \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and a scene description t into an image x depicting that scene.

To define conceptual blindspots in g_{θ} , we compare the probability mass assigned to each concept by the data-generating process with the mass induced by the model. We therefore take an evaluation set of natural image-text pairs $(\mathcal{D}_{\mathcal{X}}, \mathcal{D}_{\mathcal{T}})$ and generate a paired synthetic set $\mathcal{D}'_{\mathcal{X}}$ using the corresponding text descriptions. To estimate concept occurrence, we use an **Energy model** $\xi : \mathcal{X} \rightarrow \mathbb{R}^d$ that maps images to a d -dimensional vector whose k^{th} coordinate represents the energy of concept k , i.e., $\xi_k(x) = E(c_k)$. The invertibility assumption makes these concept-energy estimates well defined. Correspondingly, the probability assigned to image x can be written as $p_k(x) \propto \exp(-\sum_k \xi_k(x))$ (where Z_{ξ} is the partition function), yielding the population-level estimate $p_k(\mathcal{D}_{\mathcal{X}}) = \prod_{x \in \mathcal{D}_{\mathcal{X}}} p_k(x)$. Using this quantity and the sigmoid map $\sigma(\cdot)$, we define the average energy difference for concept k between $\mathcal{D}_{\mathcal{X}}$ and $\mathcal{D}'_{\mathcal{X}}$.

Definition 2 (Energy Difference). Let $x \in \mathcal{D}_{\mathcal{X}}$ denote a real image sampled from the data-generating process G , and let $x' \in \mathcal{D}'_{\mathcal{X}}$ be a synthetic image generated by the model g_{θ} . Let $\xi_k : \mathcal{X} \rightarrow \mathbb{R}$ denote the energy assigned to the k^{th} concept by the energy model ξ . We define the energy difference for concept k as:

$$\begin{aligned} \delta_{g_{\theta} \leftrightarrow G}(k) &= \sigma(\mathbb{E}_{x'}[\xi_k(x')] - \mathbb{E}_x[\xi_k(x)]) \\ &= \frac{p_k(\mathcal{D}'_{\mathcal{X}})}{p_k(\mathcal{D}_{\mathcal{X}}) + p_k(\mathcal{D}'_{\mathcal{X}})}, \end{aligned} \quad (1)$$

where the expectations are taken over $\mathcal{D}'_{\mathcal{X}}$ and $\mathcal{D}_{\mathcal{X}}$, respectively, and $p_k(\mathcal{D}) \propto \exp(-\sum_{x \in \mathcal{D}} \xi_k(x))$ denotes the unnormalized conceptual probability mass of dataset $\mathcal{D}$ under concept k .

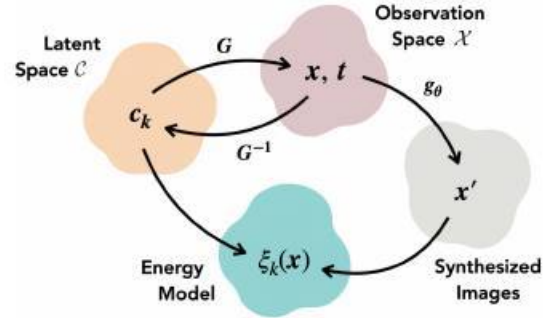


Figure 2: **Intuitive mapping of our framework.** Latent concepts $c_k \in \mathcal{C}$ map to observations (x, t) through the (invertible) DGP, while S_{θ} generates images x' from t . The energy model $\xi_k(x)$ extracts concept representations from both x and x' , enabling direct comparison of concept distributions and identification of blindspots.

Thus, the energy difference for concept k measures the relative probability that the concept occurs in generated observations ($\mathcal{D}'_{\mathcal{X}}$) compared with the natural baseline ($\mathcal{D}_{\mathcal{X}}$). This measure gives a direct definition of conceptual blindspots.

Definition 3 (Suppressed / Exaggerated Conceptual Blindspots). Given a generative image model g_{θ} , we say, compared to the data-generating process G , c_k is a suppressed conceptual blindspot in the model if $\delta_{g_{\theta} \leftrightarrow G}(k) < \lambda_{\min}$ and exaggerated if $\delta_{g_{\theta} \leftrightarrow G}(k) > \lambda_{\max}$.

Overall, a conceptual blindspot is a concept whose likelihood in generated images deviates markedly from its prevalence under the data-generating process, either through suppression or exaggeration. Suppressed concepts have disproportionately low activation (e.g., $\delta(k) < \lambda_{\min}$), whereas exaggerated concepts are overrepresented (e.g., $\delta(k) > \lambda_{\max}$). Throughout the analysis, we use $\lambda_{\min} = 0.1$ and $\lambda_{\max} = 0.9$ to isolate these high-confidence regimes.

This definition connects to “mode collapse” studied in prior work (e.g., see Bau et al. [39]), but it operates at a more precise granularity. Mode collapse concerns exaggerated or suppressed odds of generating *entire images*, whereas our analysis measures changed odds for specific concepts. For example, if a model repeatedly fails to produce images of an object with a white background, then white background is a suppressed conceptual blindspot.

4 Method: Operationalizing the Definition of Conceptual Blindspots

We now operationalize this definition as a concrete pipeline for identifying conceptual blindspots in a generative model g_{θ} . Following Sec. 3 the pipeline requires two ingredients: (i) images sampled from g_{θ} that can be compared with the ground-truth generative process, and (ii) an energy model that places natural and generated images in the same concept space. Below, we use $\|\cdot\|_F$ to denote the Frobenius norm and $\|\cdot\|_0$ to denote the number of non-zero entries (the ℓ_0 pseudo-norm). For a vector or matrix X , $X \geq 0$ implies element-wise non-negativity. For $n > 0$, we let $[n] := \{1, \dots, n\}$, and denote the i -th row of a matrix A by A_i .

From Prompts to Latent Representations. To identify conceptual blindspots in g_{θ} , we compare image-caption pairs (x, t) from $\mathcal{D}_{\mathcal{X}}$ sampled by the data-generating process G with paired synthetic counterparts produced by g_{θ} from the same text descriptions.

Given t , we synthesize a counterpart image x' with a pretrained text-to-image generator $g_{\theta} : \mathcal{T} \rightarrow \mathcal{X}$ implemented as a denoising diffusion probabilistic model (DDPM) [78, 86, 87]. Sampling proceeds in latent space through the reverse trajectory $(\gamma_t)_{t=0}^T$:

$$\gamma_T \sim \mathcal{N}(\mathbf{0}, I), \quad \gamma_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\gamma_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_{\theta}(\gamma_t, t, c) \right) + \sigma_t \eta_t, \quad \text{and} \quad \eta_t \sim \mathcal{N}(\mathbf{0}, I),$$

where $\alpha_t \in (0, 1)$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ follow the standard cosine noise schedule. A pretrained VAE decodes the final latent γ_0 into the synthetic image $x' = \text{VAE}(\gamma_0)$. For the remainder of the paper, we treat g_{θ} as a black-box prompt-to-image map $t \mapsto x'$.

Defining the Energy Model. Prior work shows that self-supervised learning methods can invert the data-generating process and identify the underlying energy function up to linear transformations [126, 128, 133, 134]. We therefore use DINOv2 [135]—a state-of-the-art self-supervised model—as the feature backbone for our analysis. Because the number of concepts underlying the DGP can exceed the dimensionality of the model’s feature space [51, 136], we train sparse autoencoders (SAEs) on DINOv2 features to recover subspaces corresponding to these concepts [50, 56, 137–139]. The key intuition is that if DINOv2 represents the concepts underlying the generative process through approximately orthogonal directions (as assumed by the independence constraint in Def. 1), then an SAE can isolate those concepts along individual latent dimensions [136]. The activation assigned by the SAE to each dimension then serves as our approximation to the ground-truth energy function for the concept represented by that dimension.

Formally, let $f : \mathcal{X} \rightarrow \mathbb{R}^d$ denote the DINOv2 feature extractor. We compute features $a = f(x) \in \mathbb{R}^d$ for both natural and synthetic images in $\mathcal{D}_{\mathcal{X}}, \mathcal{D}'_{\mathcal{X}}$. For n image-text pairs, we stack real and generated features into matrices $A, A' \in \mathbb{R}^{n \times d}$. The SAE decomposes each feature vector into a sparse combination of learned concept atoms. Let $D \in \mathbb{R}^{d \times K'}$ denote a dictionary of K'

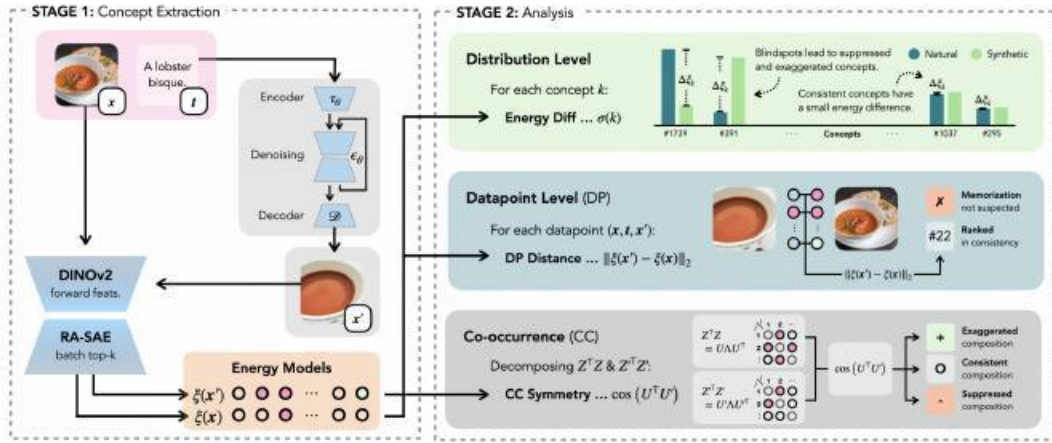


Figure 3: **Concept Extraction Pipeline.** For a triplet (x, t, x') , we extract DINOv2 features from each image and pass them through RA-SAE to obtain sparse concept embeddings, yielding energy concept vectors, and let $\Psi: \mathbb{R}^n \rightarrow \mathbb{R}^{n \times K}$ be the SAE encoder that maps input features to sparse codes. Applying Ψ row-wise yields the activation matrix $Z = \Psi(A) \in \mathbb{R}^{n \times K}$, where each row $z_i = \Psi(a_i)$ is the concept decomposition of one image. The decoder reconstructs features as ZD^T , and training minimizes reconstruction error under sparsity and non-negativity constraints:

$$\min_{\Psi, D} \|A - \Psi(A)D^T\|_F^2 \quad \text{s.t.} \quad \Psi(A) \geq 0, \|\Psi(A)_i\|_0 \ll K' \quad \forall i \in [n]. \quad (2)$$

Vanilla SAEs can drift toward arbitrarily oriented dictionaries, which makes downstream concept analyses sensitive to random initialization. To remove this instability and make the analysis reproducible, we use the *Archetypal* SAE (RA-SAE) [56] with a TOP-K sparsity constraint [137]. RA-SAE anchors the dictionary D as a convex combination of training data. Specifically, we write $D = WA$ with $W \in \Omega_{K', n}$, the set of row-stochastic matrices in $\mathbb{R}^{K' \times n}$:

$$\Omega_{K', n} := \{W \in \mathbb{R}^{K' \times n} \mid W \geq 0, W1 = 1\}. \quad (3)$$

Thus every atom D_i lies in the convex hull of the data $\text{conv}(A)$, and every reconstruction ZD^T lies inside the conic hull of the data $\text{cone}(A)$. This constraint keeps the learned concepts faithful to the support of the data distribution [56]. After training, the SAE provides a consistent sparse code basis: Z for real images and Z' for generated counterparts. *These codes represent prompt-conditioned visual semantics in a shared, interpretable concept basis, and their activation values serve as the energy values used to analyze conceptual blindspots.* In summary, the method maps a prompt and associated real image (t, x) to paired sparse concept vectors (z, z') , enabling direct comparison of real and generated visual content in a common conceptual basis.

The full pipeline therefore starts from a captioned image (t, x) , synthesizes a counterpart x' , and maps both images into a shared sparse concept space through a vision encoder and trained SAE. The resulting interpretable representations (z, z') provide the foundation for measuring conceptual shifts induced by the generative process.

5 Results

We evaluate four generative image models trained on LAION-5B—SD 1.5, SD 2.1, PixArt, and Kandinsky—using $|\mathcal{X}| = 10,000$ image-text pairs and their corresponding generations (Fig. 4). The analysis establishes blindspots at three complementary levels (Fig. 3): a **distribution-level** evaluation reveals suppressed and exaggerated concepts; a **datapoint-level** analysis isolates failures tied to ambiguity, omission, and memorization; and a **compositional** analysis exposes distortions in concept co-occurrence geometry.

The analysis is supported by an interactive exploratory tool, shown in Fig. 5. This web interface centers on a UMAP projection of con-



Figure 4: **Representative image-caption pairs** from LAION-5B and matching generations produced from the same prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

cept representations and enables direct visualization and comparison of concept-level energy differences. It is publicly available at a software repository, together with pre-computed energy-difference data for the four evaluated models (SD 1.5, SD 2.1, PixArt, and Kandinsky). The findings below are derived from the tool’s core capabilities:

- **Contrast different models and architectures.** For each evaluated model, the tool provides a UMAP visualization spanning all 32,000 RA-SAE concepts. Each scatter point is one concept, color-coded by its energy difference.
- **Inspect concepts.** Each concept card reports key statistics, representative real and generated images (x , x'), and visualized co-occurrence patterns.
- **Explore blindspots.** Beyond UMAP and per-concept views, the tool provides global rankings of suppressed and exaggerated blindspots, making the strongest conceptual failures immediately visible.

Rather than limiting the paper to one case study, we report representative findings that demonstrate the tool’s breadth and enable broader, customizable exploration.

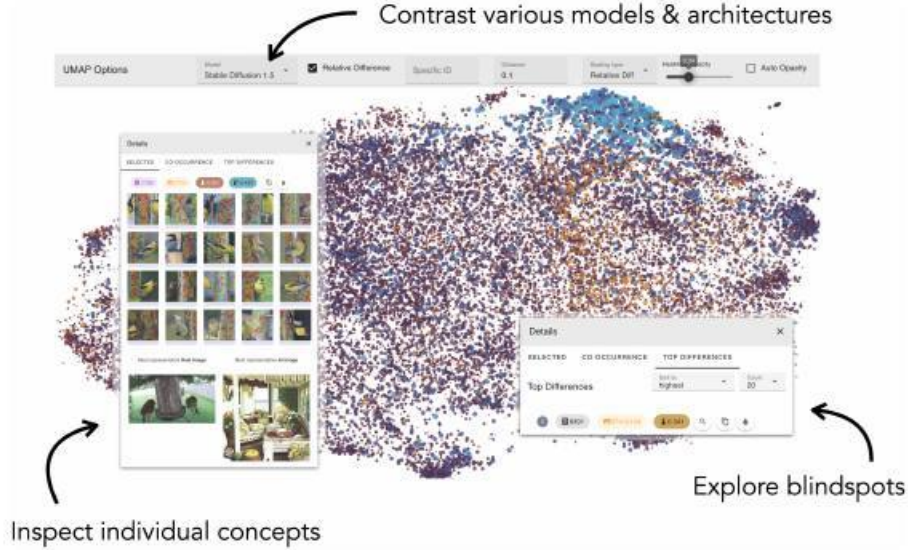


Figure 5: **Overview of the Exploratory Tool.** The web interface displays a UMAP projection for each evaluated model, where each dot is a concept color-coded by energy difference. Selecting a concept opens a detail panel with illustrative images, statistics, and the most representative natural and generated images x and x' . The interface also reports ordered co-occurrences and global rankings of blindspots.

5.1 Generative Image Models Cannot Produce Unseen Concepts

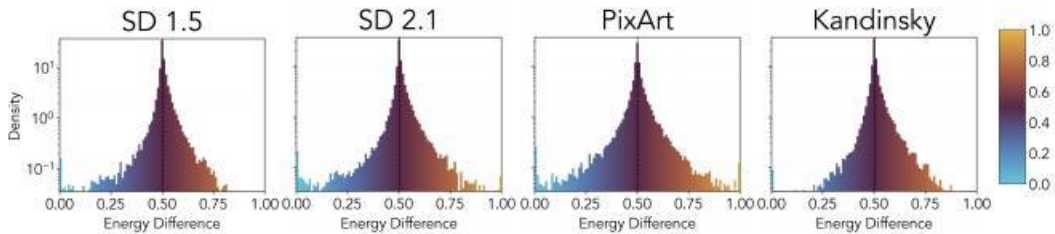


Figure 6: **Concept Energy Distribution.** Log-scale histograms of energy differences $\delta(k)$ across 32,000 concepts compare natural and synthesized distributions for each evaluated model. Values left of zero identify suppressed concepts (under-represented), while values right of zero identify exaggerated concepts (over-represented).

We first quantify disparities between each generative model and the natural image distribution by analyzing the marginal energy difference $\delta(k)$ across 32,000 RA-SAE concepts. As defined in Sec. 4, this quantity measures the relative prevalence of each concept in synthesized versus natural image sets. Values $\delta(k) < 0.1$ mark under-represented, suppressed concepts, whereas $\delta(k) > 0.9$ marks over-represented, exaggerated concepts. Fig. 6 reports the $\delta(k)$ distribution for each evaluated model. Across all models, the histograms are heavy-tailed and place substantial mass at both extremes, demonstrating systematic discrepancies in concept coverage. The left tail, corresponding to suppressed concepts, is consistently denser and longer than the right tail, showing that concept suppression is the dominant imbalance. This asymmetry is captured by the negative skewness of the distributions: Skewness = -0.54 for SD 2.1, -0.40 for both SD 1.5 and PixArt, and -0.23 for Kandinsky.

All models exhibit both suppressed and exaggerated concepts, but their profiles differ substantially. PixArt, for instance, shows a wider spread, indicating a more strongly suppressed concept distribution. The consistent left-skew across all distributions establishes concept omission as a common failure pattern, while the model-specific profiles motivate the more detailed analyses that follow.

5.2 Structure and Specificity of Conceptual Blindspots

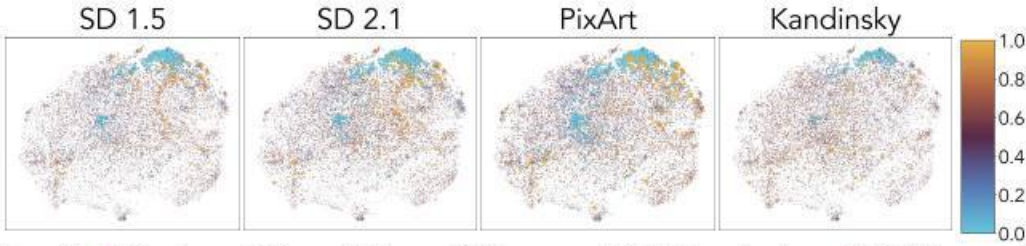


Figure 7: **Structure of Concept Energy Differences.** UMAP visualizations of 32,000 concepts, colored by energy difference $\delta(k)$ between natural and synthesized distributions. Clustered regions expose structured conceptual blindspots, with suppressed concepts on the blue end and exaggerated concepts on the yellow end.

Having quantified marginal concept-frequency discrepancies, we now show that these discrepancies have global structure. We embed all 32,000 concepts into two dimensions using UMAP on the sparse codes and color each concept by its $\delta(\cdot)$ value. Fig. 7 reveals distinct concept clusters across all models. These clusters frequently form contiguous regions of conceptual blindspots, especially for suppressed (blue) concepts, showing that blindspots are structured rather than randomly scattered. This structure points to shared biases in training distributions, architectural priors, or both. To measure how consistently these blindspot patterns transfer across models, we analyze both the magnitude and structure of concept-level $\delta(\cdot)$ values. Fig. 8 presents scatter plots and pairwise Pearson correlations between the $\delta(k)$ vectors of SD 1.5 and the other evaluated models. SD 1.5 and 2.1 exhibit strong correlation ($r = 0.82$), reflecting their shared architectural and training pipelines. In contrast, correlations with PixArt and Kandinsky are much lower— $r = 0.41$ and $r = 0.46$, respectively—showing that different architectures emphasize different regions of the conceptual space.

Overall, this analysis demonstrates that some blindspots are shared across models—likely reflecting dataset-level properties—whereas others are strongly model-specific and reflect training dynamics, architecture, or model capacity. This motivates a direct examination of both universal blindspots and blindspots unique to individual models.

5.3 Qualitative Examples of Blindspots

We next visualize concrete suppressed and exaggerated blindspots to show what these regimes capture. Fig. 9a shows a conceptual blindspot suppressed by all models: every evaluated model fails to reproduce the concept *solid white on documents*. Although the caption explicitly references this concept, none of the generated images realizes the intended visual semantics, providing direct evidence that this region of concept space is systematically under-sampled across models. Fig. 9b highlights a model-specific blindspot: the concept *pan* is captured by three models but is

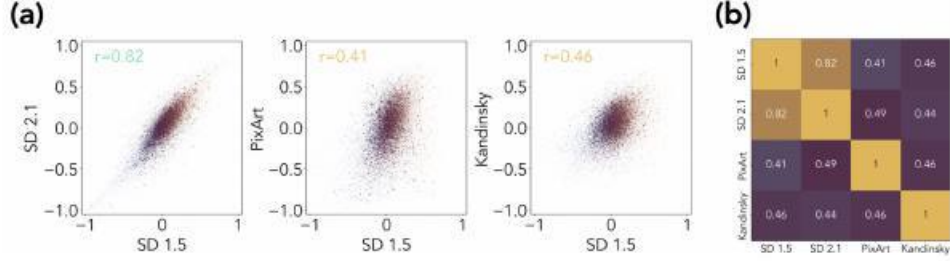


Figure 8: **Cross-Model Concept Energy Correlation.** Pairwise scatter plots of $\delta(k)$ across all four evaluated models, with Pearson correlation coefficients reported top left. Strong alignment between SD 1.5 and SD 2.1 contrasts with weaker correlations among other architectures, exposing model-specific blindspots. **Correlation Matrix of Conceptual Blindspots.** Heatmap of pairwise Pearson correlation coefficients for δ between all models, quantifying shared conceptual blindspots across these models.

conspicuously absent from Kandinsky generations. This example reinforces Sec. 5.2: cross-model agreement is strong for some blindspots but decisively limited for others.

Conversely, Fig. 10 shows an exaggerated blindspot: generated images overemphasize the concept shadow under animal. Although shadows are plausible, their consistent and pronounced rendering across models, relative to the more nuanced and variable shadows in natural images, indicates an overactive prior. Notably, despite searching for concepts uniquely exaggerated by a single model, we did not find clear examples; in this analysis, exaggerated blindspots appear largely shared across models.

Together, these examples demonstrate that conceptual blindspots manifest visibly in generated outputs and that our energy-based diagnostic surfaces both shared and model-specific failure modes. The consistent failure to reproduce clear or solid background elements such as whitespaces on documents is especially revealing: it identifies a limitation that cuts across all evaluated models and may reflect architectural constraints, training-data biases, or both.

These aggregate-level analyses already establish systematic concept failures, but they also motivate a sharper question: do blindspots appear only across populations, or do they manifest at individual data-



Figure 10: **Example of an Exaggerated Conceptual Blindspot.** Four synthesized images D'_x , archetypal for the concept shadow under animal, are shown alongside the corresponding natural image D_x and caption t used to generate

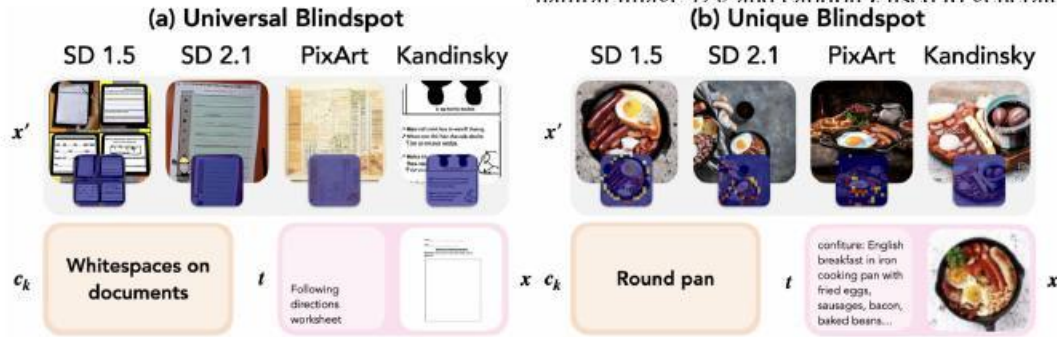


Figure 9: **Examples of Suppressed Conceptual Blindspots.** Natural images x , representative c_k , and t are shown alongside four synthesized images x' generated using S_θ . The universal blindspot appears in all evaluated models; the unique blindspot appears only in Kandinsky.

points? This finer-grained
perspective probes mecha-
nisms directly and uncov-
ers prompt misinterpreta-
tion, latent memorization, or
both.

5.4 Investigating Datapoint-level Energy Difference: From Incongruent to Memorized Images

To move beyond population-level statistics, we examine individual natural-generated image pairs with the largest and smallest concept-averaged $\delta(\cdot)$ differences. This analysis makes model successes and failures interpretable at the sample level; the failures often trace to prompt ambiguity or memorization artifacts. Fig. 11a shows instances with near-zero differences in average $\delta(\cdot)$ values. In these cases, the generated images are conceptually nearly indistinguishable from the originals. However, qualitative inspection shows that this alignment does not necessarily reflect faithful prompt understanding; it often reflects replication of memorized templates. Repeated visual structures, such as clothing outlines or object arrangements, indicate that the model may copy highly frequent training patterns. By contrast, Fig. 11b shows samples among the largest $\delta(\cdot)$ values. These samples exhibit substantial conceptual divergence between synthesized and natural images. Some discrepancies arise from underspecified or noisy captions, but others reveal genuine blindspots: the prompt describes a clear concept present in $\mathcal{D}_X$, yet the model fails to realize it in $\mathcal{D}'_X$. These cases show that even when language grounding is adequate, certain concepts remain outside the model’s effective generative repertoire.



Figure 11: **Datapoint-level Conceptual Alignment.** (a) Minimal-energy-difference examples where models appear to memorize training patterns. (b) Large-difference examples where prompt ambiguity or model limitations produce substantial concept divergence.

373 5.5 Analyzing Post-Training Effects

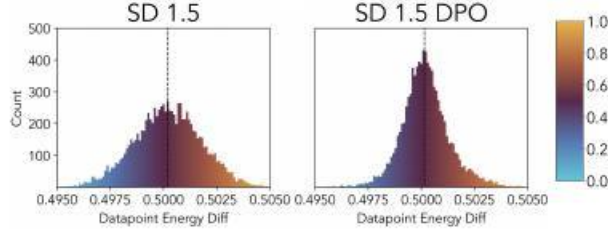


Figure 12: **Effect of DPO on Concept Fidelity.** Histogram of datapoint-wise energy differences between synthesized and natural distributions for SD 1.5 models with and without DPO.

374 Post-training protocols, including safety fine-tuning, can reduce the diversity of model genera-
 375 tions [140]. Because our pipeline isolates differences between a model’s generations and the ground-
 376 truth DGP, we use it to characterize the effects of DPO—a popular safety fine-tuning protocol [141].
 377 We compare two SD 1.5 variants: one trained with DPO and one without it. For each image
 378 pair $(\mathcal{D}_X, \mathcal{D}'_X)$, we compute the ℓ_2 norm of the difference between their concept energy vectors,
 379 $\|\xi(\mathcal{D}'_X) - \xi(\mathcal{D}_X)\|_2$. Fig. 12 presents the resulting datapoint-wise energy-difference histogram. The
 380 DPO-enhanced model has both a lower median and a narrower spread, indicating stronger and more
 381 consistent alignment of generated concepts with the ground-truth DGP. This evidence indicates that
 382 DPO regularizes the model’s concept distribution, encouraging outputs that more closely reflect the
 383 semantic content of the paired natural inputs. Although our analysis does not isolate the specific
 384 inductive biases introduced by DPO, the results provide empirical evidence that its human-preference
 385 objective indirectly improves matching to the training distribution. In particular, it reduces *both*
 386 *over and under* activation of individual concepts relative to the natural baseline. These findings
 387 demonstrate that our pipeline characterizes post-training interventions not merely by output quality,
 388 but by how they reshape the conceptual geometry of the model’s output space.

389 5.6 Conceptual Misalignment as a Function of Empirical Frequency

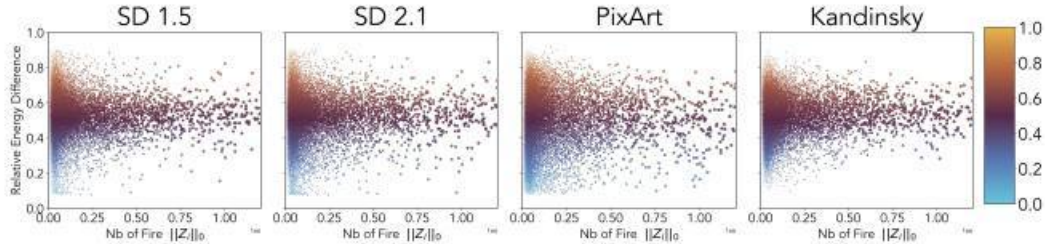


Figure 13: **Concept Fidelity Across Frequency Spectrum.** Scatter plots show the relationship between concept frequency (x-axis) and energy difference (y-axis) across four evaluated models. Each point is a concept, with size proportional to activation frequency.

390 We hypothesize that conceptual blindspots are not merely architectural artifacts; they can also emerge
 391 directly from distributional properties of specific features. We test this hypothesis by examining
 392 whether concepts rarely activated in natural images—the long tail of the data distribution—are more
 393 likely to become blindspots in generative models. We process the natural dataset $\mathcal{D}_X$ through the
 394 trained SAE and compute, for each concept k , its empirical frequency $\|Z_{:,k}\|_0$, where $Z_{:,k}$ contains
 395 activations of concept k across all images. We then correlate this frequency with the absolute energy
 396 difference observed in generated outputs. Fig. 13 visualizes this relationship for all evaluated models.
 397 Concepts with higher natural-data frequency tend to show lower energy discrepancies, while rare
 398 concepts—especially suppressed ones ($\delta(k) < 0.5$)—show substantial alignment errors. This pattern
 399 supports the conclusion that many blindspots arise from systematic long-tail effects rather than
 400 random quirks. Addressing them therefore likely requires not only architectural changes but also
 401 data-centric interventions such as reweighting or augmentation.

6 Discussion

Our analysis exposes multiple conceptual blindspots in four widely used generative image models and demonstrates that these failures are measurable, structured, and interpretable. The results here are deliberately broad rather than exhaustive: each finding could support a dedicated investigation, but together they establish the versatility of the method and exploratory tool. Out of the box, the framework identifies concepts that models struggle to generate, detects memorization artifacts, discovers datapoints with insufficient captions, quantifies post-training effects, and characterizes conceptual shifts across model architectures. This creates a clear path for follow-up work to deepen causal analysis, expand the range of evaluated architectures, and investigate the roots of conceptual blindspots. Future work can also use hierarchical RA-SAE concept representations to support more nuanced analysis.

Limitations. The framework has important boundaries. Because concept extraction relies on DINOv2 and RA-SAE, the analysis is constrained to concepts these models represent; concepts poorly captured by them will remain outside our measurement. In addition, although 10,000 images provide a substantial evaluation set, this sample may still under-cover the extreme long tail of rare concepts, concept co-occurrence patterns, and other compositional statistics.

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Appendix

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A Experimental Setup

This section specifies the experimental setup used to evaluate four widely used generative image models: SD 1.5/2.1, Kandinsky, and PixArt, all trained on LAION-5B or its subsets/derivatives. The code is available at a software repository.

A.1 Observation Space

We construct the observation space by sampling 10,000 image-text pairs from LAION-5B [93], which serves as the natural-image domain for our comparisons. Because concerns about CSAM and other unsafe content made the original release unavailable, we use a filtered substitute subset available at <https://huggingface.co/datasets/laion/relaion2B-en-research-safe>

The sampling procedure has three steps: (1) load the LAION dataset using the Hugging Face `datasets` library, (2) validate URL structure and resource availability with HTTP HEAD requests, and (3) randomly sample with replacement until obtaining 10,000 valid samples. This produces D_G with accessible (x, t) tuples, giving a reliable comparison set despite the substantial unavailability of image URLs from the original release. Additional synthesized-image examples appear in Appendix

H

A.2 Synthesized Images

For each evaluated model, we generate a synthetic dataset D_{g_θ} in one-to-one correspondence with D_G , yielding triplets (x, x', t) . For each of the 10,000 image-text pairs (x, t) in D_G , we use t to synthesize a counterpart image x' with each generative model g_θ .

Synthesis follows the standard text-to-image pipeline for each architecture, implemented with the Hugging Face `diffusers` library and mixed-precision model loading (fp16). All synthetic images are generated at 512×512 resolution using default parameters.

Stable Diffusion 1.5. We use the checkpoint from <https://huggingface.co/benjamin-paine/stable-diffusion-v1-5> a mirror of the deprecated <https://huggingface.co/runwayml/stable-diffusion-v1-5>. Inference uses 50 steps with guidance scale fixed at 7.5.

Stable Diffusion 1.5 + DPO. The DPO variant of SD 1.5 used in Sec. 5.5 follows the baseline SD 1.5 implementation but replaces the UNet with the DPO-trained version from <https://huggingface.co/mhdang/dpo-sd1.5-text2image-v1>

Stable Diffusion 2.1. We use the checkpoint from <https://huggingface.co/stabilityai/stable-diffusion-2-1>. Inference uses 50 steps with guidance scale fixed at 7.5.

Kandinsky. We use the checkpoint from <https://huggingface.co/kandinsky-community/kandinsky-2-1>. Inference uses 100 steps with guidance scale fixed at 4.0.

PixArt. We use the checkpoint from <https://huggingface.co/PixArt-alpha/PixArt-XL-2-1024-MS>. Inference uses 50 steps with guidance scale fixed at 7.5.

A.3 Distribution Level Analysis

Section 5.1 We compute energy differences $\delta(\cdot)$ across all 32,000 concepts for each evaluated model. A sigmoid transformation with temperature $T = 0.8$ normalizes the scores, and the resulting values are visualized as log-scale density histograms with 100 bins spanning $[0, 1]$.

Section 5.2 We embed all 32,000 concepts into two-dimensional space using UMAP on the sparse concept codes. Each point is an individual concept colored by energy difference $\delta(\cdot)$, making suppressed and exaggerated blindspots visible. To quantify cross-model consistency, we compute pairwise Pearson correlations between $\delta(\cdot)$ vectors for all model pairs and report both scatter plots and correlation matrices. This analysis determines whether blindspots cluster in concept space and separates model-specific from universal blindspot patterns.

891 **Section 5.3** We rank all 32,000 concepts by energy difference $\delta(\cdot)$ and manually inspect the
 892 extrema, covering both suppressed and exaggerated blindspots. For suppressed blindspots, we select
 893 concepts with $\delta(\cdot) < 0.1$; for exaggerated blindspots, we choose those with $\delta(\cdot) < 0.9$. The presented
 894 examples are manually annotated with textual descriptions based on their most activating images and
 895 spatial attention patterns. Appendix D describes our efforts to automate this interpretation step.

896 **Section 5.5** We compare SD 1.5 with and without DPO as follows: for each image pair (x, x') , we
 897 compute the L2 norm of the difference between their concept energy vectors $\|\xi(x') - \xi(x)\|_2$. We
 898 apply a sigmoid transformation with temperature $T = 0.8$ to the element-wise differences before
 899 taking their mean. This produces datapoint-wise energy differences that quantify how far each
 900 generated image deviates from its natural counterpart in concept space. We visualize these differences
 901 as overlapping histograms to directly contrast the two variants.

902 **Section 5.6** For each concept c_k , we count its empirical frequency $\|\mathbf{Z}_{:,i}\|_0$, the number of non-zero
 903 activations across the natural dataset. We then apply sigmoid normalization with temperature $T = 0.4$
 904 to the energy differences $\delta(\cdot)$. Scatter plots visualize the analysis, with empirical concept frequency
 905 on the x-axis and sigmoid-transformed energy difference on the y-axis. Point size is proportional to
 906 activation frequency, and point color is proportional to the magnitude of energy differences.

907 A.4 • Datapoint Level Analysis

908 **Section 5.4** For each image pair (x, x') , we compute the L2 norm of the difference between
 909 their concept energy vectors $\|\xi(x') - \xi(x)\|_2$. This yields a scalar conceptual-divergence score for
 910 each pair. We rank samples by this score: minimal divergence identifies potential memorization
 911 artifacts, while maximal divergence identifies strong conceptual failures. This analysis enables
 912 targeted qualitative inspection of specific failure modes.

913 A.5 • Co-occurrence Analysis

914 **Appendix F** For both natural and synthesized data D_G and D_{g_θ} , we analyze concept co-occurrence
 915 through the co-activation matrix $Z^T Z$, which stores pairwise correlations in concept usage. Spectral
 916 analysis via eigendecomposition identifies dominant conceptual directions. We assess alignment
 917 between natural and synthetic co-occurrence structures with cosine-similarity heatmaps between
 918 the top-100 eigenvectors of each co-occurrence matrix. These 100×100 similarity matrices are
 919 visualized as square heatmaps: diagonal alignment indicates matching principal concept axes, while
 920 off-diagonal structure reveals rotations and mismatches in compositional geometry.

921 B Custom RA-SAE Training Details

922 This section specifies the training details of our custom archetypal SAE (RA-SAE) [56]. The model
923 is open-sourced at a software repository.

924 **Dataset.** We train the auto-encoder on the complete ImageNet-1k training split (≈ 1.28 M RGB
925 images). DINOv2 [135] converts each image into 261 visual tokens, which are fed to the SAE without
926 class or position embeddings. The total number of training tokens is therefore $50 \times 1.28\text{M} \times 261 \approx$
927 1.67×10^{10} .

928 **Dictionary.** The dictionary contains 32,000 concept dimensions. The sparse activation rule uses
929 top- k masking with $k = 5$, setting all activations outside the largest five per input to 0. We initialize
930 weights with Xavier/Glorot, train at mixed precision (fp16), and run the final ten epochs at full
931 precision.

932 **Optimizer and Schedule.** We train for 50 epochs with AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$) and weight
933 decay 10^{-5} . The schedule applies linear warm-up over the first 5% of steps, followed by cosine
934 decay from $\eta_{\max} = 5 \times 10^{-4}$ to $\eta_{\text{final}} = 10^{-6}$. The objective combines MSE loss with an auxiliary
935 penalty for activations that never enter the top- k set, with $\lambda = 10^{-5}$.

936 C Computational Resources

937 This section reports the GPU resources used for training and experiments. In total, the study used
938 approximately 202 GPU-hours on NVIDIA H100s and H200s.

939 **RA-SAE.** Training required approximately 24 GPU-hours on three NVIDIA H100s.

940 **Synthesized Images.** Generating the full $D_{g\theta}$ (see Appendix A) took roughly 5 hours per generator
941 when distributed across four NVIDIA H200 GPUs. Across five generators, this totaled approximately
942 100 GPU-hours on NVIDIA H200s.

943 **● Distribution-Level Analysis.** Extracting distribution-level energy differences took about 3 hours
944 per generator on a single NVIDIA H200 GPU (total ~ 15 GPU-hours).

945 **● Datapoint-Level Analysis.** Computing datapoint-level energy differences, ranking concepts per
946 datapoint, and ranking datapoints per concept took approximately 3 hours per generator on one H200
947 GPU (total ~ 15 GPU-hours).

948 D Autointerpretability

949 To automate interpretation of RA-SAE concepts, we qualitatively evaluated whether Vision-Language
 950 Models (VLMs) can describe them. We queried ChatGPT 4o through the OpenAI API with multiple
 951 prompts and image formats for an image x'' in which the target concept was salient, recording the
 952 VLM-generated description. This section reports the main findings. For evaluation, we allowed the
 953 VLM to provide full reasoning; a scalable deployment would simply request only the final description
 954 and omit the reasoning.

955 First, we provided the VLM with two images: the raw x'' and a heatmap-superimposed version of
 956 x'' that localized concept intensity:

Input



You are given two images: (1) raw image and (2) the same image with a heatmap of areas that are salient for a concept present in the image, as identified by a DINOv2-based SAE. Describe what is this concept.

Output

The concept identified by the DINOv2-based SAE in these images appears to correspond to **"ring"** or **"metal ring"** as the salient object.

Reasoning:

- In the original image, a sleek, brushed-metal ring is centered on a wooden surface.
- In the heatmap overlay, the regions that are most activated (least transparent with bright color blocks) are tightly focused on the ring itself, particularly the outer surface and inner reflections.
- The wooden background is mostly ignored except for minor scattered activations, confirming that the concept is not about the background texture.
- The consistent and strong focus on the metallic, circular object suggests that the model recognizes this as the defining element—likely trained to detect the shape, material, and reflectivity typical of rings.

957 The VLM did not interpret the heatmap correctly. Instead of describing the wood texture in the
 958 background—the actual salient region—it described the foreground ring, which received no attention
 959 from the RA-SAE.

960 To test whether the failure was caused by insufficient prompting, we repeated the scenario with an
 961 expanded prompt that explained the heatmap in greater detail:

Input



You are given two images: (1) raw image and (2) the same image with a superimposed heatmap of areas that contain a concept identified by a DINOv2-based SAE. Red areas are the most salient (contain the concept); blue areas are the least salient (do not contain the concept). Describe what is this concept.

Output

The concept identified by the DINOv2-based SAE in these images appears to be **"ring"** or more specifically **"metal band/jewelry ring."**

Here's the reasoning:

- In the heatmap image, the red and yellow (most salient) regions align very closely with the circular metal ring at the center of the image.
- The ring is the only distinct object in the photo and is clearly segmented from the background (wooden surface), which is mostly marked in blue and green (least salient).
- The attention does not focus on the texture of the background or lighting but is highly localized to the curved, metallic object.

Thus, the concept is almost certainly the metallic ring—its shape, reflectivity, and material seem to trigger the sparse autoencoder's activation, making it a learned concept in the SAE dictionary.

962 This prompt modification did not resolve the issue: the VLM again identified the ring as the prominent
 963 concept. Additional prompting strategies, including more detailed heatmap explanations and inclusion
 964 of the heatmap-generation code, produced similar failures.

965 We next provided only the heatmap-superimposed x'' together with a similar prompt explaining how
966 to interpret the heatmap:

Input	Output
 You are given an image with a superimposed heatmap of areas that contain a concept identified by a DINOv2-based SAE. Red areas are the most salient (contain the concept); blue areas are the least salient (do not contain the concept). Describe what is this concept.	The concept identified by the DINOv2-based SAE in this image appears to be "ring" or more broadly, "circular metallic object". The most salient (red) areas are focused tightly around the ring itself—especially along its edges and reflective interior—indicating that the model has strongly associated those regions with the concept. The surrounding wooden surface is mostly in blue and green, which shows it is not relevant to the concept, supporting the idea that the model is isolating the ring as the key object of interest.

967 The result again failed to identify the wooden background as the prominent concept and instead
968 focused on the ring. Additional prompting strategies with this image format left the outcome largely
969 unchanged.

970 These observations show that the heatmap alone is insufficient to direct the VLM to the salient area;
971 irrelevant regions continue to distract the model. We therefore replaced the heatmap-superimposed
972 x'' with an alpha-masked version of x'' :

Input	Output
 You are given an image where only areas that contain a concept identified by a DINOv2-based SAE are preserved; areas without the concept were removed. Describe what is this concept.	The concept identified and preserved in this image appears to be "wood texture" or "wood grain." The retained areas show consistent patterns typical of wooden surfaces—linear, fibrous streaks in earthy tones like beige, gray, and brown. The selective masking highlights parts of the image where this natural texture is most prominent, suggesting the SAE learned to isolate the visual features of wood (likely flooring or a table).

973 With the alpha-masked input, the VLM correctly identified wood texture as the dominant concept.
974 We observed consistent success across a range of concepts and images.

975 The alpha-masked formatting of x'' and the associated prompt yielded the most reliable results in our
976 qualitative evaluation. The evaluation remains limited because it is qualitative, lacks ground-truth
977 concept annotations, and uses a single VLM; nevertheless, it identifies a clear and practical direction
978 for scalable autointerpretability of SAE concepts.

979 E Additional Results: Model-Specific Blindspots

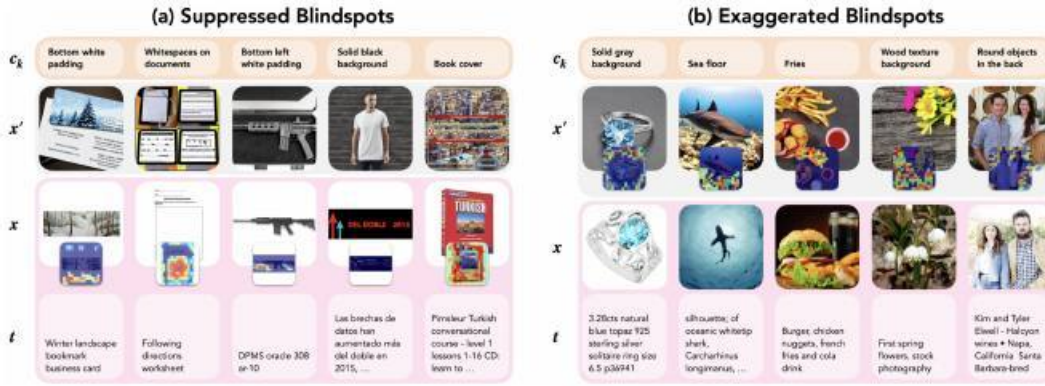


Figure 14: Conceptual blindspots in **Stable Diffusion 1.5**. For each concept, we show the prototypical natural image (for suppressed blindspots) or synthesized image (for exaggerated blindspots) selected by highest absolute activation. The concept heatmap is superimposed on the image.

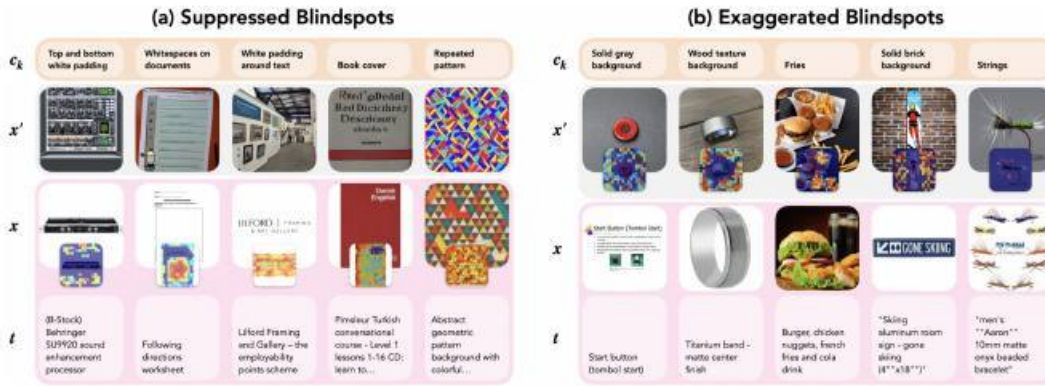


Figure 15: Conceptual blindspots in **Stable Diffusion 2.1**. For each concept, we show the prototypical natural image (for suppressed blindspots) or synthesized image (for exaggerated blindspots) selected by highest absolute activation. The concept heatmap is superimposed on the image.

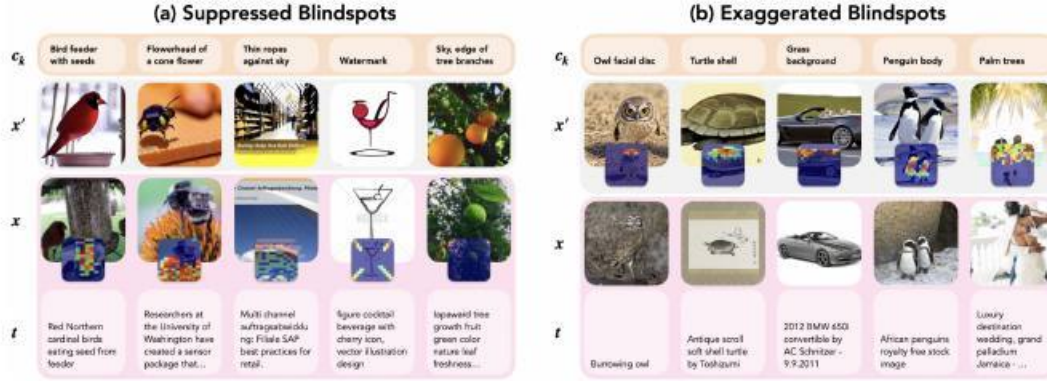


Figure 16: Conceptual blindspots in **Kandinsky**. For each concept, we show the prototypical natural image (for suppressed blindspots) or synthesized image (for exaggerated blindspots) selected by highest absolute activation. The concept heatmap is superimposed on the image.



Figure 17: Conceptual blindspots in **PixArt**. For each concept, we show the prototypical natural image (for suppressed blindspots) or synthesized image (for exaggerated blindspots) selected by highest absolute activation. The concept heatmap is superimposed on the image.

F Additional Results: Higher-order Blindspots with Compositional Discrepancy

Thus far, the analysis has centered on individual concept activations. Visual scenes, however, are not collections of isolated concepts; they are organized through rich co-occurrence patterns that encode compositional semantics. We therefore test whether generative models preserve this higher-order structure by analyzing the co-activation matrix $Z^T Z$, which reflects pairwise correlations in concept usage.

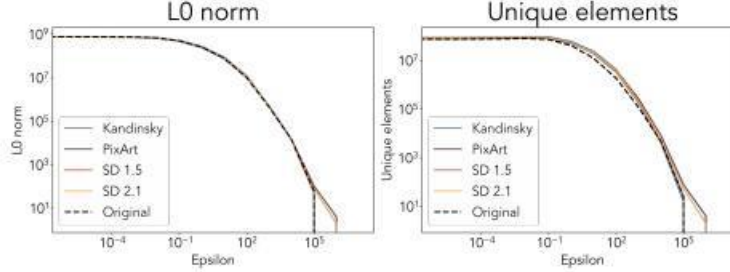


Figure 18: • **Sparsity and Structural Divergence.** Left: L0 norm of the co-occurrence matrix $Z Z^T$ as a function of ϵ (threshold), showing how many entries remain active in each model. Right: Number of unique entries in the synthesized distribution relative to the natural distribution. All evaluated models preserve global sparsity structure but diverge in activation content.

At the level of binary structure, diffusion models approximate the global sparsity of the natural co-occurrence matrix with high fidelity. As shown in Fig. 18 (left), the ℓ_0 norm of $Z^T Z$ —thresholded at varying ϵ values—tracks closely between natural and synthesized distributions across all models. This shows that the gross connectivity of the conceptual graph, i.e., which concepts co-activate at all, is well preserved. Formally, $Z^T Z$ can be interpreted as the adjacency matrix of a weighted, undirected graph over concepts, where edge weights measure co-activation strength across the dataset.

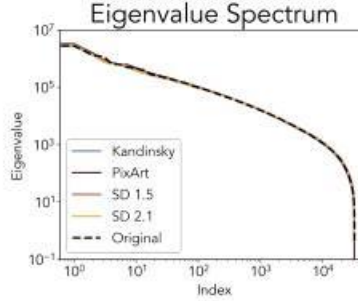


Figure 19: • **Spectral Structure of Co-occurrence.** Log-log plot of eigenvalue spectra from co-occurrence matrices $Z Z^T$ across models and the natural distribution. All evaluated models match the heavy-tailed decay of the natural distribution.

However, Fig. 18 (right) shows that the specific content of these co-activations diverges: a substantial fraction of entries in model-generated $Z^T Z$ are absent from the natural baseline. Thus, models retain broad compositional capacity while shifting the identity of active concept pairings, potentially reflecting model-specific inductive biases or training artifacts. To probe the internal structure of these co-occurrence patterns, we use spectral analysis. Fig. 19 shows eigenvalue spectra of the co-occurrence matrices for each model and the natural distribution. All spectra exhibit heavy-tailed decay consistent with power-law behavior, indicating that generative models preserve the overall rank structure and variance allocation across conceptual dimensions.

We then examine alignment of dominant conceptual directions using cosine-similarity heatmaps between the top 100 eigenvectors of synthesized and natural co-occurrence matrices (Fig. 20). All models show partial diagonal alignment, implying overlap in principal concept axes, but the off-

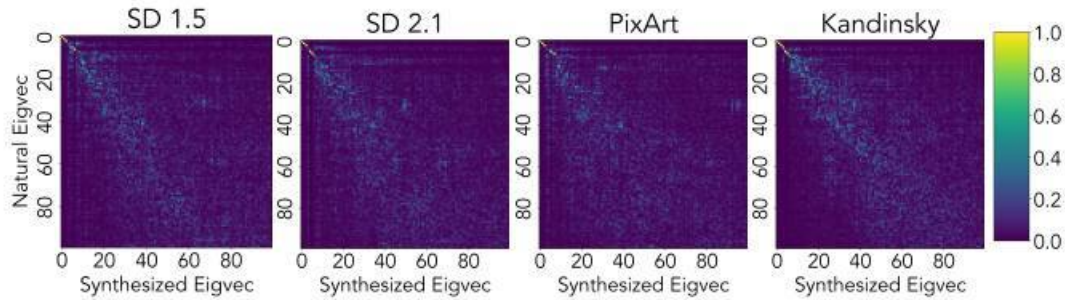


Figure 20: ● **Concept Basis Similarity.** Cosine similarity heatmaps between the top 100 eigenvectors of the natural and synthesized co-occurrence matrices ZZ^T . Diagonal structure shows alignment of dominant conceptual directions, with alignment strength varying across the four models.

1005 diagonal entries reveal rotations and mismatches in higher modes. These patterns establish structured
 1006 deviations in compositional geometry.

1007 Together, these findings show that diffusion models approximate the global shape of concept co-
 1008 activation well, yet deviate in subtle and structured ways under spectral analysis. These higher-order
 1009 discrepancies provide a plausible mechanism for failures in coherent multi-object scenes or relational
 1010 concepts.

1011 G Additional Results: Blindspot Stress Testing

1012 To stress-test the blindspots identified by our method, we collected prompts describing each target
 1013 blindspot and generated many images from them. We then contrasted outputs from models in which
 1014 the concept was identified as a blindspot with outputs from models in which it was not.

1015 Specifically, ChatGPT-4o was prompted as follows: *I want to generate an image of the following*
 1016 *concept: "<blindspot>". Suggest 50 prompts highlighting this concept to be used as input for a*
 1017 *text-to-image model. Return these as a list of strings in Python.* We generated five images per prompt
 1018 and analyzed them with our custom RA-SAE model (see Appendix B), which ranked images by the
 1019 intensity of the desired concept. We then manually reviewed all images to determine whether the
 1020 blindspot was successfully depicted.

1021 The suppressed-concept examples confirm the diagnostic. Some partial attributes occasionally appear,
 1022 such as a holder or string for the bird feeder blindspot and a round hole for the glossy DVD
 1023 disc blindspot, but the models generally fail to generate the full concept. This behavior matches the
 1024 method’s assessment and validates the stress-test protocol.

1025 G.1 Bird Feeder Blindspot in Kandinsky



Figure 21: Stress-test images generated from prompts involving the bird feeder concept. Kandinsky, where our method identifies this concept as a suppressed conceptual blindspot, fails to generate a corresponding image. By contrast, SD 1.5, where this concept is not identified as a blindspot, successfully generates the concept.



Figure 22: Stress-test images generated from prompts involving the glossy DVD disc concept. SD 1.5, where our method identifies this concept as a suppressed conceptual blindspot, fails to generate a corresponding image. By contrast, Kandinsky, where this concept is not identified as a blindspot, successfully generates the concept.

1027 **H Additional Examples: Synthesized Images**



Figure 23: Additional LAION-5B image-caption examples with matching generations produced from the same prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

1028 I Concentration bounds for δ

1029 In our experiments, we estimate $\delta(k)$ using $n = 10,000$ paired samples for each concept. While this
 1030 budget is modest, it raises the natural question of whether it suffices to obtain reliable estimates. To
 1031 address this, we derive a concentration bound on the empirical estimator $\hat{\delta}_n(k)$ using McDiarmid’s
 1032 inequality [142]. The resulting bound is tight and demonstrates that even with relatively few samples,
 1033 we can obtain fast and accurate estimates of concept bias.

1034 **Theorem 4** (Concentration of $\hat{\delta}_n(k)$). *We assume that the concept score $\xi_k(\mathbf{x})$ takes values in $[a, b]$
 1035 almost surely for all images $\mathbf{x}$ drawn from either $\mathcal{D}_X$ or $\mathcal{D}'_X$. Let n paired samples $(\mathbf{x}_i, \mathbf{x}'_i)_{i=1}^n$ be
 1036 drawn independently with $\mathbf{x}_i \sim \mathcal{D}_X$ and $\mathbf{x}'_i \sim \mathcal{D}'_X$, and define the empirical estimator*

$$\hat{\delta}_n(k) := \sigma \left(\frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}_i) - \frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}'_i) \right).$$

1037 Let $M := b - a$ and $L := M/4$. Then for every $\varepsilon > 0$, the deviation satisfies

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right).$$

1038 *Proof.* The function $x \mapsto \sigma(x)$ is $1/4$ -Lipschitz, since $|\sigma'(x)| \leq 1/4$ for all x .

1039 Viewing $\hat{\delta}_n(k)$ as a function of the $2n$ independent variables $(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}'_1, \dots, \mathbf{x}'_n)$, changing a
 1040 single argument alters the inner difference of means by at most M/n , and the outer sigmoid scales this
 1041 by at most $1/4$. Hence, the bounded difference constant for each coordinate is $(M/n)(1/4) = L/n$.

1042 By McDiarmid’s inequality [142],

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2\varepsilon^2}{\sum_{j=1}^{2n} (L/n)^2} \right) = 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right),$$

1043 which proves the claim. □

1044 In practice, most concept scores $\delta(k)$ are sparse, with the majority concentrated near zero and only
 1045 a few reaching values up to 10. The concentration bound shows that even for the largest observed
 1046 biases, a sample size of $n = 10,000$ yields estimates of $\hat{\delta}_n(k)$ that deviate from the true value by at
 1047 most a small ε with high probability. This directly supports our sampling strategy and confirms that
 1048 accurate bias measurements are attainable with limited data.

1049 J Monotonicity and Calibration-Free Interpretation of δ

1050 When analyzing blindspots, the central objective is to rank concepts by the severity of their generative
 1051 bias. We use $\delta(k)$ for this purpose. The result below establishes that $\delta(k)$ is a strictly increasing
 1052 reparameterization of both the energy gap and the odds ratio, so it preserves their ordering exactly.
 1053 Thus, no calibration is needed when using $\delta(k)$ to rank concepts.

1054 **Theorem 5** (Monotonicity and Calibration of $\delta_{g_\theta \leftrightarrow G}$). *For every concept index k define the energy*
 1055 *gap*

$$\Delta_k = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}'_{\mathcal{X}}}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{\mathcal{X}}}[\xi_k(\mathbf{x})],$$

1056 *the associated odds ratio $\rho_k = \exp(\Delta_k)$, and the energy–difference score*

$$\delta(k) = \frac{1}{1 + \exp(-\Delta_k)} = \frac{\rho_k}{1 + \rho_k}.$$

1057 *Then $\delta(k)$ is a strictly increasing bijection of both Δ_k and ρ_k , so ranking concepts by any one of*
 1058 *$\delta(k)$, Δ_k , or ρ_k produces exactly the same ordering.*

1059 *Proof.* The logistic sigmoid satisfies $\sigma'(x) = \sigma(x)(1 - \sigma(x)) > 0, \forall x \in \mathbb{R}$; hence σ and therefore
 1060 $\delta(k) = \sigma(\Delta_k)$ grow strictly with Δ_k . Because the exponential map is also strictly increasing and
 1061 bijective $\mathbb{R} \rightarrow (0, \infty)$, setting $\rho_k = \exp(\Delta_k)$ preserves order and gives $\Delta_k = \log \rho_k$. Substituting
 1062 this identity into σ yields $\delta(k) = \sigma(\log \rho_k) = \rho_k / (1 + \rho_k)$, which is the composition of two
 1063 strictly increasing bijections and is therefore itself strictly increasing and bijective in ρ_k . Since strict
 1064 monotonic functions never reverse inequalities, the three quantities share the same total order over
 1065 concepts.

1066 □

1067 Thus, ranking concepts by $\delta(\cdot)$ is strictly equivalent to ranking them by energy gap or by conceptual
 1068 generation odds ρ_k . No calibration is necessary: all three quantities induce the same total ordering
 1069 over concepts.

K Stability of FID Under SAE Embeddings

In this section we establish a quantitative relationship between the Fréchet Inception Distance (FID) computed in the original activation space (of dimension d) and the FID after applying a (potentially overcomplete) SAE dictionary $D \in \mathbb{R}^{k \times d}$ with $k \gg d$. Throughout we assume that D has orthonormal columns but is not necessarily square, i.e.

$$D^\top D = I_d, \quad \text{while} \quad DD^\top \neq I_k.$$

We start by recalling a simple fact: if D is not overcomplete, orthogonal and $k = d$, then we have an isometry between A and Z , implying that the FID is perfectly preserved. However, this case is not realistic, we will then turn to the overcomplete case, and show we can bound FID by the extreme singular value of D . We will work with the Wasserstein-2 metric $\mathcal{W}_2$, noting that FID is just $\mathcal{W}_2^2$ specialised to Gaussians.

For a probability measure μ on $\mathbb{R}^d$ we write $D_\# \mu$ for its push-forward under D , i.e. $D_\# \mu(z) = \mu(D^{-1}z)$. Denote by $\sigma_{\min}$ and $\sigma_{\max}$ the minimal and maximal singular values of D , equivalently the square-roots of the extremal eigenvalues of $DD^\top$:

$$\sigma_{\min}^2 I_k \preceq DD^\top \preceq \sigma_{\max}^2 I_k.$$

Empirically one usually finds $\sigma_{\min}, \sigma_{\max} \approx 1$, but the proof does not rely on that. We will start by a simple lemma in the case where D is not overcomplete.

Lemma 1 (Isometry under exact orthogonality). *Suppose $k = d$ and $D^\top D = DD^\top = I_d$. Then D is an isometry: $\|Dv\|_2 = \|v\|_2$ for all $v \in \mathbb{R}^d$. Consequently, for any probability measures μ, ν on $\mathbb{R}^d$ with finite second moment,*

$$\mathcal{W}_2(D_\# \mu, D_\# \nu) = \mathcal{W}_2(\mu, \nu).$$

Proof. Orthogonality of D implies preservation of the Euclidean norm, and push-forward commutes with the map inside the $\mathcal{W}_2$ infimum; the integrand is unchanged, so the infimum value is identical. $\square$

This case, however, is quite unrealistic as SAE usually rely on the overcompleteness to extract meaningful and interpretable concepts. In the overcomplete case, D is no longer orthonormal, but we can still have column-orthonormal dictionary. We will use that to show that we can bound using the extremal singular value of $D^\top D$.

Theorem 6 (FID under column orthogonal embeddings). *Let $D \in \mathbb{R}^{k \times d}$ satisfy $D^\top D = I_d$ and denote by $0 < \sigma_{\min} \leq \sigma_{\max}$ the extreme singular values of $DD^\top$. Given two data matrices $A, A' \in \mathbb{R}^{n \times d}$ (rows are sample vectors), set $Z = AD^\top \in \mathbb{R}^{n \times k}$ and $Z' = A'D^\top \in \mathbb{R}^{n \times k}$. Then*

$$\sigma_{\min}^2 \text{FID}(A, A') \leq \text{FID}(Z, Z') \leq \sigma_{\max}^2 \text{FID}(A, A').$$

Proof. Let μ denote the empirical measure of A and ν the empirical measure of A' , i.e.

$$\mu = \frac{1}{n} \sum_{i=1}^n \delta_{A_{i,:}}, \quad \nu = \frac{1}{n} \sum_{i=1}^n \delta_{A'_{i,:}}.$$

For any coupling $\pi \in \Pi(\mu, \nu)$ (i.e. a probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ, ν) we have, by the extremal singular value bound,

$$\sigma_{\min}^2 \|x - y\|_2^2 \leq \|D(x - y)\|_2^2 \leq \sigma_{\max}^2 \|x - y\|_2^2, \quad \forall (x, y) \in \mathbb{R}^d \times \mathbb{R}^d.$$

Integrating with respect to any coupling $\pi \in \Pi(\mu, \nu)$ yields

$$\sigma_{\min}^2 \int \|x - y\|_2^2 d\pi \leq \int \|D(x - y)\|_2^2 d\pi \leq \sigma_{\max}^2 \int \|x - y\|_2^2 d\pi.$$

The middle integral is exactly the transport cost of the pushed-forward coupling $(D \times D)_\# \pi$ between $\mu_D := D_\# \mu$ and $\nu_D := D_\# \nu$. Because the inequalities hold for every π , they also hold for

1104 the optimal couplings attaining $\mathcal{W}_2(\boldsymbol{\mu}, \boldsymbol{\nu})$ and $\mathcal{W}_2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D)$, although these two optima need not
 1105 coincide. Taking the infimum over $\boldsymbol{\pi}$ term-wise makes this explicit:

$$\sigma_{\min}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi} \leq \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|D(\mathbf{x} - \mathbf{y})\|_2^2 d\boldsymbol{\pi} \leq \sigma_{\max}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi}.$$

1106 Hence

$$\sigma_{\min}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}) \leq \mathcal{W}_2^2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D) \leq \sigma_{\max}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

1107 Recognizing $\text{FID}(\cdot, \cdot) = \mathcal{W}_2^2(\cdot, \cdot)$ for the Gaussian surrogate and substituting $(\mathbf{A}, \mathbf{A}')$ (respectively
 1108 $(\mathbf{Z}, \mathbf{Z}')$) proves the result. $\square$

1109 Theorem 6 establishes that applying a column-orthogonal overcomplete SAE dictionary cannot distort
 1110 Fréchet Inception Distance by more than the square of its extremal singular values. When $\mathbf{D}$ is nearly
 1111 orthogonal—as we empirically found with $\sigma_{\min}, \sigma_{\max} \approx 1$ —FID measured in the SAE feature space
 1112 remains close to canonical FID. $\square$

1113